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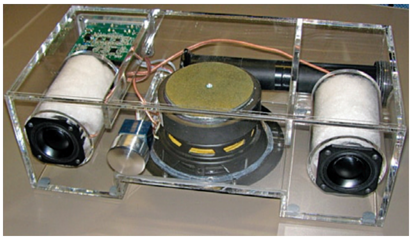
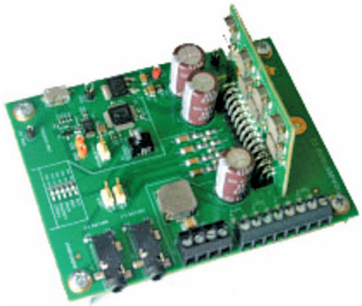
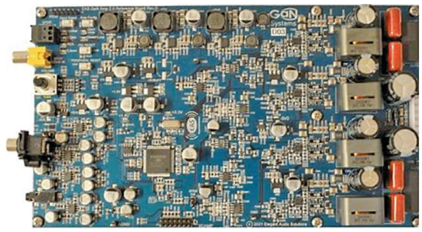
A 2.1-Channel Audio Amplifier	Four-Channel Audio Amplifier	GaN Class-D Audio Amplifier
		
The 2.1-channel Class-D audio amplifier features a design with high-efficiency architecture and active EQ for bass equalisation. The setup delivers sound quality from a small enclosure.	The Class-D amplifier mimics a Class-AB amplifier for easier evaluation, solving thermal and noise issues. It is optimised for compact spaces, robust enough for harsh automotive conditions, and meets CISPR-25 standards for electromagnetic interference, ensuring reliable performance.	This Class-D audio amplifier is a high-performance platform delivering up to 400 watts with minimal power loss, featuring over 96% efficiency, ultra-low THD+N, and versatile input support.
The Class-D, 2.1-channel audio amplifier reference design by Analog Devices is housed in a single, finely calibrated box enclosure that includes all electronic and speaker components, requiring only an external power supply and signal source to function. This system comprises two 5cm loudspeakers for the stereo channels and a 12.5cm loudspeaker for the subwoofer, powered by two MAX98400A amplifiers—one in stereo mode for the left and right channels and another in mono for the subwoofer. The design is split into electronic circuitry and speaker/enclosure assembly. Notable features include its compact, all-in-one format operating on a 12V to 20V DC power supply, achieving high sound pressure levels from a small enclosure, utilising high-efficiency Class-D architecture, and incorporating active EQ with dynamic bass equalisation for enhanced sound quality. The input stage handles both stereo and subwoofer signals, with volume controlled by a stereo, log-taper digital potentiometer and differential sensing to avoid ground loops. The same MAX98400A model also drives the subwoofer in mono mode, delivering 44W into 4Ω by paralleling two Class-D outputs.	The TIDA-00573 reference design from Texas Instruments makes it easy to evaluate while reducing thermal issues and minimising switching noises characteristic of Class-D amplifiers. This design comprises an audio amplifier module and the necessary circuitry for integration into a base board, enabling stand-alone performance testing. It is ideal for automotive applications such as colour media or display audio, mid- and low-range head units, and seat vibration audio amplifier modules, enhancing the multimedia experience in vehicles. The design is optimised for space and robust enough to handle load-dump conditions and start/stop operations down to 5.6 volts, and meets CISPR-25 standards for electromagnetic interference, ensuring reliability in demanding environments. The system is intended to serve as a form, fit, and functional evaluation platform, emulating the typical configuration of Class-AB amplifiers.	The GaN Class-D audio amplifier reference design by Infineon, model GS-EVB-AUD-AMP2-GS, is an advanced platform for developing high-performance audio amplifiers, demonstrating the capabilities of a stereo Class-D audio amplifier optimised for robust sound quality with minimal power loss. The board outputs 200 watts per channel into an 8-ohm load and 300 watts into a 4-ohm load, with the ability to continuously handle up to 400 watts, making it suitable for a variety of audio applications. Efficiency exceeds 96% under full load due to the integration of GaN E-HEMTs, and the THD+N is less than 0.03%, ensuring clear audio. Engineered for fanless operation and requiring minimal external components due to the high level of integration with the D2Audio controller/digital signal processor (DSP), the amplifier supports both open-loop and closed-loop operations and accommodates various audio inputs, making it highly adaptable for integration into different audio systems.
The reference design suits personal audio systems, desktop audio setups, and compact home theatres.	The reference design is suitable for various automotive audio applications.	The reference design is suitable for home theatre systems and professional audio setups.
Analog Devices (ADI)	Texas Instruments (TI)	Infineon Technologies
https://www.electronicsforu.com/electronics-projects/reference-design-for-a-2-1-channel-audio-amplifier	https://www.electronicsforu.com/electronics-projects/class-d-amplifier-reference-design	https://www.electronicsforu.com/electronics-projects/reference-design-for-gan-class-d-audio-amplifier